PROJECT 15 · ENTERPRISE SYSTEMS

Enterprise Knowledge Mining System

A document processing system using RAG, semantic search, and NLP to unlock enterprise knowledge at scale.

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PART 01

The Problem

THE PROBLEM

Extract value from unstructured enterprise documents.

Build a document processing system that extracts information from documents, enables a semantic search engine, and answers natural language questions using RAG (Retrieval-Augemented Generation).

WHAT

Ingest and process 1K-10K documents by cleaning and chunking text, extracting key entities and generating embeddings to build semantic representations for efficient search and retrieval.

WHY IT MATTERS

Every enterprise has terabytes of untapped documents. Semantic search and Q&A turns raw text into competitive intelligence.

It enables faster decision-making by reducing manual searching

Goal: Build a production-ready RAG system that unlocks value from unstructured research documents.

THE PROBLEM

Extract value from unstructured enterprise documents.

Build a document processing system that extracts information from documents, enables a semantic search engine, and answers natural language questions using RAG (Retrieval-Augemented Generation).

WHAT MAKES IT HARD

Specialized domain language, inconsistent document formats, large-scale datasets, and the need for low-latency responses with accurate, source-backed citations for real-time Q&A.

Goal: Build a production-ready RAG system that unlocks value from unstructured research documents.

PART 02

The Data

A freely available, curated document corpus of research articles.

arXiv Research Papers Subset

arXiv API – Free

- ~1060 papers
- PDF versions used for each paper
- Query restricted to only Computer Science articles
- PDFs contained different types and formats of information (e.g. Raw Text, Split Columns, Images, Formulae)

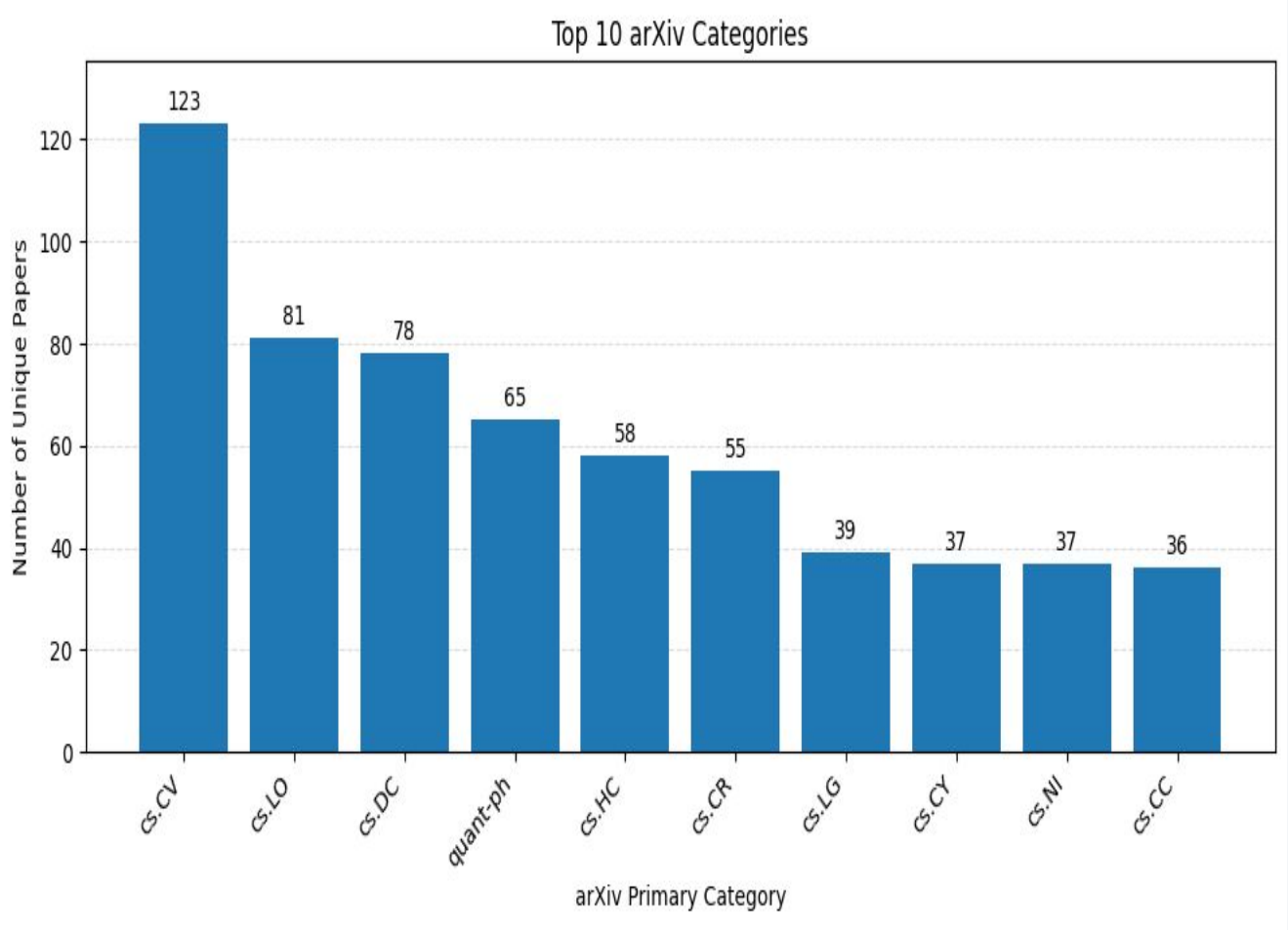
DATA PROBLEMS

PDF links in arXiv response provide no document

```
  Download failed 0505043v2: HTTP Error 404: Not Found
  Download failed 2206.01573v3: HTTP Error 404: Not Found
  Download failed 1310.0319v3: HTTP Error 404: Not Found
Processing papers...
  FAILED 1310.0319v3: HTTP Error 404: Not Found
  FAILED 0505043v2: HTTP Error 404: Not Found
```

Rate Limitations from the arXiv API

```
Fetching papers 401-500...
429 rate-limit, waiting 6s...
429 rate-limit, waiting 12s...
429 rate-limit, waiting 24s...
```



Dataset Overview

- Dominated by **Computer Vision (cs.CV)**, followed by **Logic (cs.LO)** and **Distributed Systems (cs.DC)**
- Also includes diverse areas: **Quantum Physics, HCI, Cryptography & Security, Machine Learning, Computers & Society, Networking & Internet Architecture, and Computer Complexity.**

Key Insight

- The system will perform **strongest in high-volume areas** (CV, Logic, Distributed Systems)
- Still supports a **broad range of technical and interdisciplinary topics**

PART 03

Methodology

PIPELINE

From raw documents to a live Q&A.

1	2	3	4	5	6
Ingest	**Extract**	**Embed**	**Index**	**Retrieve**	**Answer**
Load & parse PDF documents	Cleaning, Section extraction, Chunking and NER	OpenAI	ChromaDB (vector store)	Semantic search	OpenAI (RAG response generation)

TOOLS PER STAGE

PyMuPDF	LangChain Recursive Character Text Splitter, & en_core_web_sm	text-embedding-3-small	ChromaDB	Custom semantic search (Chroma + cosine similarity)	OpenAI LLM gpt-4.1-mini

TECHNICAL REQUIREMENTS

Tools Used.

Document Processing

PyMuPDF (fitz), LangChain

- PDF parsing to plain text.
- Cleaning (remove artifacts, page markers, hyphenation etc).
- Section-aware chunking.

NLP

spaCy 3.8.11 (en_core_web_sm)

- Named Entity Recognition (NER) on cleaned text.
- Entity extraction for metadata enrichment.
- Noise reduction (exclude titles/abstracts from NER).

EMBEDDINGS

OpenAI (text-embedding-3-small)

- Converts text chunks into vector representations.
- Embedding generation for semantic search.
- Captures contextual meaning of text.

TECHNICAL REQUIREMENTS

Tools Used.

Vector Database

ChromaDB (PersistentClient)

- Persistent vector storage (local disk).
-ChromaDB search metric: cosine distance (evaluation score we used cosine similarity)
- Stores embeddings + metadata for retrieval.

Semantic Search

Custom (ChromaDB + cosine similarity + entity overlap)

- Dense retrieval using vector similarity.
- Hybrid re-ranking using entity overlap scoring.
- Top-k retrieval for RAG context.

RAG

OpenAI (gpt-4.1-mini)

- Retrieval-augmented generation using semantic search engine.
- Context assembly from top-k retrieved chunks.
- Answer generation grounded in retrieved context with source-aware prompting.

PART 04

Live Demo

Streamlit UI

PART 05

Evaluation & Results

PERFORMANCE METRICS

Results from notebook testing across retrieval, NER, and RAG generation on 1,057 arXiv papers.

Retrieval Precision

Top-5 recall

Semantic search passed all 5 test queries. Top cosine similarity ranged 0.57–0.76 across query types. Avg similarity at k=5 was 0.578.

Answer Accuracy

LLM Q&A Score

RAG answered all 4 test cases correctly. Answerable queries returned grounded responses; unrelated query ("chocolate cake") correctly returned no relevant context.

Hybrid Reranking

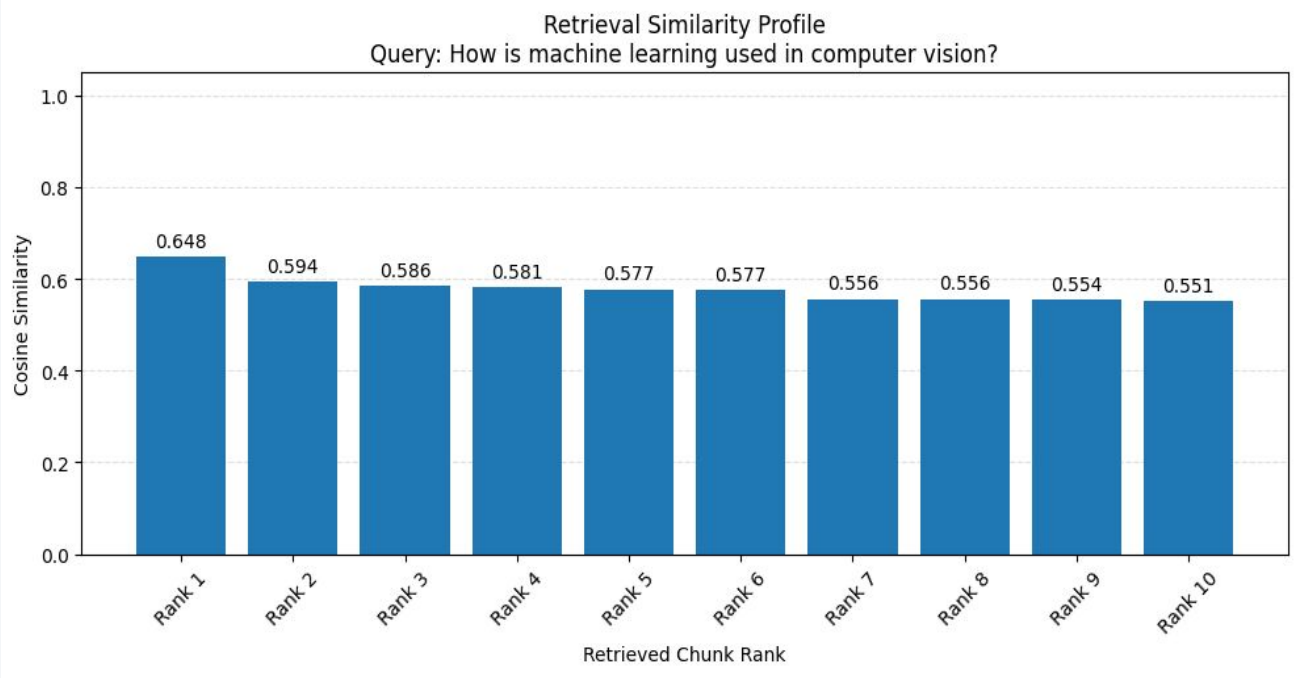
1/5 changed rank

1 of 5 hybrid queries changed top result vs. semantic. Entity overlap was 0 for most queries ; hybrid behaves near-identically to pure semantic search in practice.

DATA ANALYSIS

Retrieval & RAG Source Confidence

Retrieval Similarity Profile

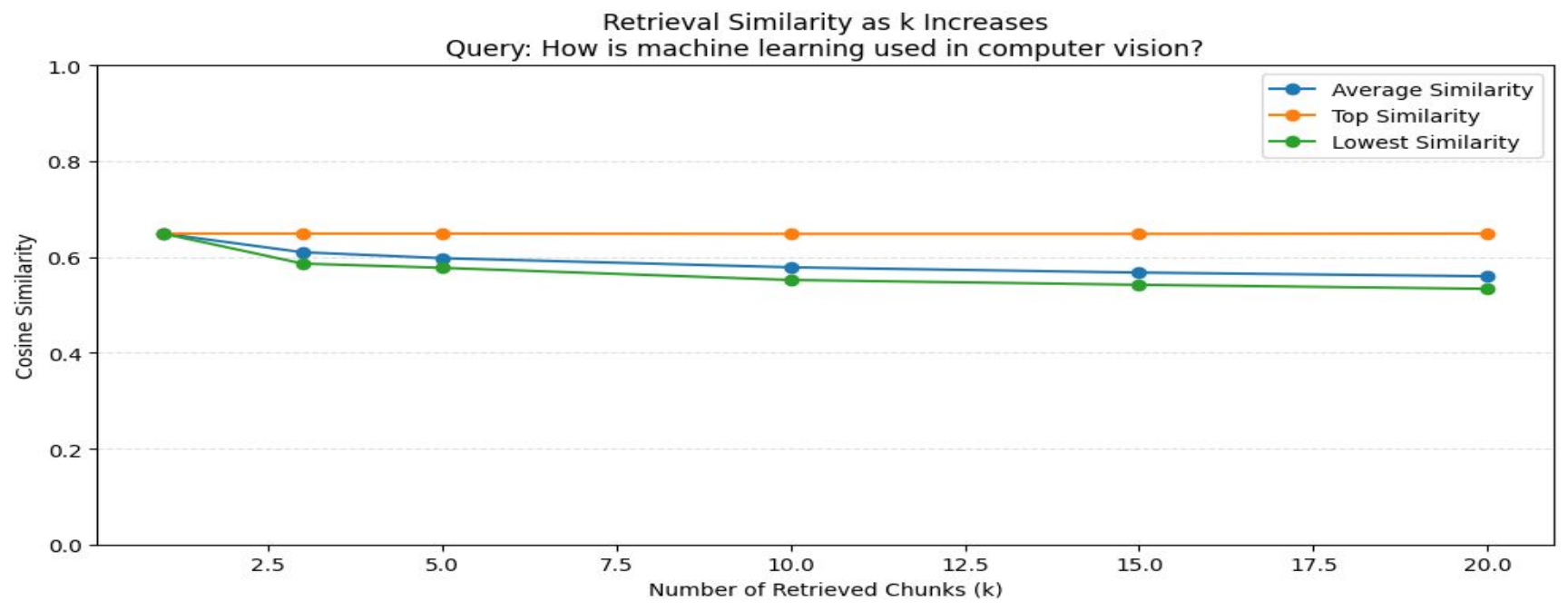


Top similarity 0.648; scores decrease gradually across ranks, confirming semantic ranking is working.

DATA ANALYSIS

Retrieval Similarity as k Increases

Query: “How is machine learning used in computer vision?”



Top similarity stays flat at 0.648 regardless of k — best chunk is always found

Average drops 0.648 → 0.560 from k=1 to k=20 as weaker chunks are included

k=5 is the sweet spot — enough context while avg similarity stays above 0.597

PERFORMANCE METRICS

Results from notebook testing across retrieval, NER, and RAG generation on 1,057 arXiv papers.

Category Filtering

5/5 passed

All 5 category filter tests passed. Retrieved chunks matched the requested arXiv category in every case, confirming correct metadata filtering.

Entity Coverage

spaCy NER F1

Top entities found: CPU, GPU, IEEE, Turing, CNN. However, figure references like "Fig" and "- 1" were also tagged as entities — PDF extraction noise that slipped through NER unfiltered.

Chunk Quality

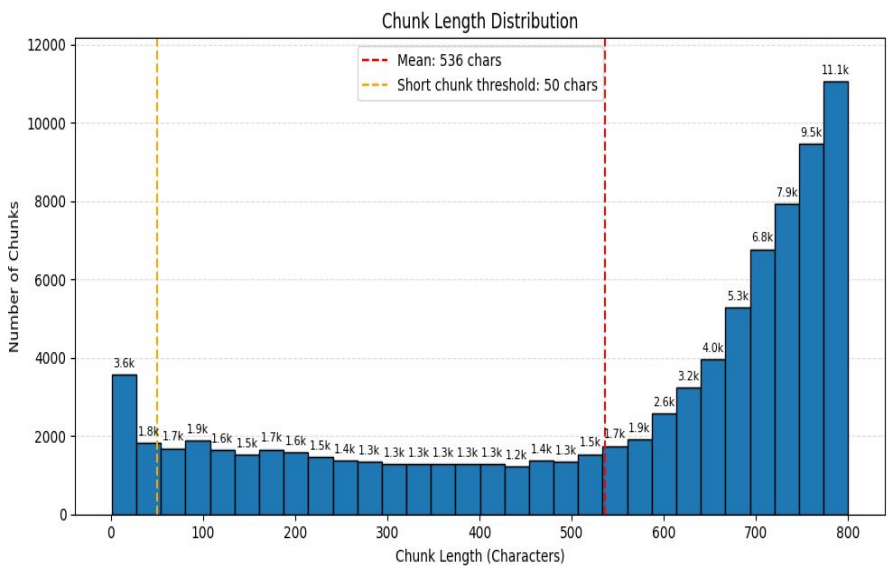
85K vectors

85,422 chunks stored across 1,057 papers. Avg chunk length 536 chars; max 800 chars. Only 5.89% of chunks under 50 chars (noise from PDF artifacts).

DATA ANALYSIS

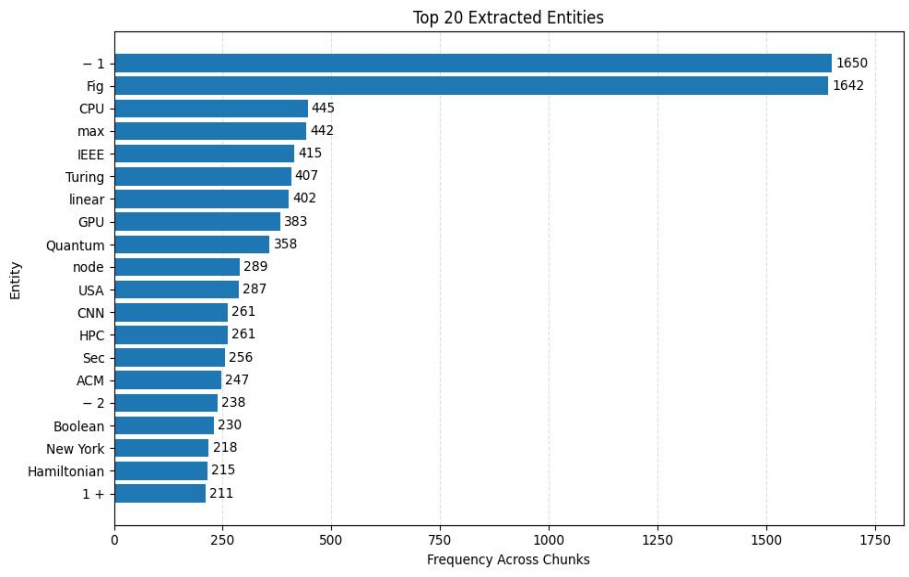
Corpus Composition & Chunk Quality

Chunk Length Distribution



Most chunks cluster near the 800-char max. Only 5.89% fall below the 50-char noise threshold.

Top 20 Extracted Entities



Technical terms (CPU, GPU, Turing) dominate. Noisy artifacts ("Fig", "- 1") reveal PDF extraction bleed-through.

PART 06

Conclusion & Future Work

HONEST LIMITATIONS + WHAT WAS LEARNED

What the current system does not handle well.

Build a system that processes documents, enables semantic search, and answers natural language questions using RAG.

NO MULTI-MODAL SUPPORT

The pipeline processes text-only PDF content. Tables, figures, and diagrams are not extracted or analyzed.

Content is parsed using PyMuPDF, so information not well represented as text may be partially lost.

NER BLIND SPOTS

spaCy NER does not consistently capture domain-specific research terms (e.g., “transformer”, “embedding”) as entities.

As a result, entity overlap is often low, limiting the effectiveness of hybrid re-ranking beyond pure semantic similarity.

RETRIEVAL DROP-OFF

The more results retrieved, the weaker the matches tend to become. Vague questions return loosely relevant chunks, while specific questions stay on target.

The system uses fixed top-k retrieval with no adaptive cutoff when results become less relevant.

FUTURE PLANS

Planned enhancements.

Feedback Loop

Collect user feedback (e.g., thumbs up/down) on responses to evaluate answer quality and guide future improvements to retrieval and prompting.

Graph Knowledge Layer

Incorporate a Neo4j-based graph layer to link related entities across documents, enabling richer relationships and improved multi-hop reasoning.

Multi-Modal Ingestion

Extend the pipeline to process tables, figures, and images, improving coverage beyond text-only content.

THANK YOU

Questions?

https://github.com/Enterprise-Knowledge-Mining-System/Enterprise-Knowledge-Mining-System

https://enterprise-knowledge-mining-system-enterpr-streamlit-app-qwxgr8.streamlit.app